

1. Symptom icons (wheeze/cough/dyspnea/tightness)

WHAT YOU SEE

A row of athlete-badge icons at the stadium gate — wheeze, cough, dyspnea, chest tightness — the four 'players' of clinical suspicion that get you into the stadium but can't win the game without objective testing.

WHAT IT ENCODES

The clinical suspicion of asthma rests on the quartet of variable symptoms — expiratory wheeze, cough (often nocturnal), dyspnea, and chest tightness — that fluctuate in intensity and over time, are often triggered by allergens/exercise/cold/viral infection, and frequently improve with bronchodilators. More than one symptom type and worsening at night/early morning increase the likelihood.

BOARD TRAP

Symptoms alone do NOT diagnose asthma — board answer always requires objective demonstration of variable expiratory airflow limitation. A confident clinical picture still needs spirometry/PEF/challenge confirmation.

2. Worse at night / diurnal variation

WHAT YOU SEE

A moon icon 'worse at night' and 'diurnal PEF >20%'.

WHAT IT ENCODES

Asthma classically worsens at night and in the early morning, reflecting circadian airway tone. Average diurnal peak-expiratory-flow (PEF) variability >10% in adults (>13% in children), or day-to-day variability >20% over 1–2 weeks, is diagnostic of the variable airflow limitation that defines asthma.

BOARD TRAP

Excessive PEF variability is a valid alternative objective criterion when spirometry is unavailable or non-diagnostic — but a single normal PEF/spirometry does NOT exclude asthma because airflow is variable by definition; repeat or provoke.

3. PD20 ≤ threshold = positive

WHAT YOU SEE

A scoreboard 'PD20 ≤ 400 µg = POSITIVE' (methacholine challenge).

WHAT IT ENCODES

In a methacholine bronchoprovocation test, the provocative dose/concentration causing a 20% fall in FEV1 (PD20/PC20) defines airway hyperresponsiveness. A low threshold (e.g., PC20 <4 mg/mL or PD20 below ~400 µg) is positive and supports asthma; this test is used when spirometry is normal but suspicion is high.

BOARD TRAP

Methacholine challenge has high sensitivity but modest specificity, so its main value is its NEGATIVE predictive value — a clearly negative test in a symptomatic untreated patient makes asthma unlikely and should prompt a search for mimics.

4. Check before severe (mimics/comorbid)

WHAT YOU SEE

A sign 'check before severe: sinus, obesity, GERD, OSA, VCD, adherence, technique'.

WHAT IT ENCODES

Before labeling asthma 'severe' or 'refractory,' confirm the diagnosis and address the modifiable drivers of poor control: inhaler TECHNIQUE, adherence, ongoing trigger/allergen and smoke exposure, and comorbidities — chronic rhinosinusitis, obesity, GERD, obstructive sleep apnea, vocal-cord dysfunction, and anxiety. Difficult-to-treat asthma is far more common than truly severe asthma.

BOARD TRAP

Apparent 'uncontrolled severe asthma' is most often poor inhaler technique, non-adherence, or an untreated comorbidity/VCD — escalating biologics before fixing these is the wrong step.

5. Asthma definition banner

WHAT YOU SEE

A banner 'ASTHMA = symptoms + objective variable obstruction'.

WHAT IT ENCODES

GINA defines asthma as a heterogeneous disease characterized by chronic airway inflammation, defined by a history of respiratory symptoms (wheeze, dyspnea, chest tightness, cough) that vary over time and in intensity, TOGETHER WITH variable expiratory airflow limitation. Both halves are required for diagnosis.

6. Methacholine pool (provocation)

WHAT YOU SEE

A 'METHACHOLINE POOL' with 'do not provoke if FEV <50%, MI/stroke, pregnancy, aortic aneurysm'.

WHAT IT ENCODES

Bronchoprovocation (methacholine, mannitol, or exercise challenge) is reserved for patients with suggestive symptoms but normal/non-diagnostic spirometry and no demonstrated reversibility. Methacholine directly stimulates airway muscarinic receptors to provoke a controlled FEV1 drop.

BOARD TRAP

Contraindicated with severe baseline obstruction (FEV1 <50% predicted or <1 L), recent MI or stroke (within ~3 months), uncontrolled hypertension, and known aortic aneurysm — provoking bronchospasm or the Valsalva of coughing is dangerous in these patients.

7. Bronchodilator reversibility (≥12% & ≥200mL)

WHAT YOU SEE

A 'BRONCHODILATOR' cannon 'FEV up ≥12% AND ≥200 mL'.

WHAT IT ENCODES

The single most useful confirmatory test: spirometry showing obstruction (reduced FEV1/FVC) that REVERSES after a SABA (e.g., 200–400 µg salbutamol) — defined in adults as an FEV1 increase of BOTH ≥12% AND ≥200 mL. This documents the variable airflow limitation central to the diagnosis.

BOARD TRAP

Reversibility distinguishes asthma from fixed COPD — but a NEGATIVE reversibility test does not exclude asthma (patient may be well-controlled or already on ICS); repeat off bronchodilators, use PEF variability, or proceed to challenge testing.

8. Eosinophils ≥ 300 = type 2

WHAT YOU SEE

A lab panel on the scoreboard reading 'eosinophils $\geq 300/\mu\text{L}$ = type 2' — the eosinophil count is the biomarker that flags the type-2 phenotype and predicts ICS / anti-IL-5 biologic response.

WHAT IT ENCODES

Blood eosinophils ≥ 300 cells/ μL (some thresholds ≥ 150) flag the type-2/eosinophilic phenotype and predict response to ICS and to anti-IL-5/anti-IL-4R α biologics. Eosinophilia is one of the key biomarkers (with FeNO and IgE) used to phenotype severe asthma for targeted therapy.

BOARD TRAP

Markedly high eosinophilia ($>1500/\mu\text{L}$) plus sinusitis, neuropathy, and other organ involvement should raise EGPA rather than plain asthma — don't over-attribute extreme eosinophilia to ordinary asthma.

9. FeNO / IgE biomarkers

WHAT YOU SEE

Panels 'FeNO untreated $>30\text{--}40$ ppb' and 'IgE $>1000 \rightarrow$ ABPA'.

WHAT IT ENCODES

Fractional exhaled nitric oxide (FeNO) reflects IL-13-driven type-2 airway inflammation; in untreated adults $>40\text{--}50$ ppb is high (intermediate 25–50) and predicts ICS responsiveness. Total IgE supports allergic phenotype and is used to dose omalizumab; very high IgE (often >1000 IU/mL) with central bronchiectasis and Aspergillus sensitization suggests ABPA.

BOARD TRAP

FeNO and IgE phenotype the patient and guide biologic choice — they do NOT establish the diagnosis or replace lung-function testing, and FeNO is suppressed by ICS and smoking (causing false-low values).

10. Silent chest = impending failure

WHAT YOU SEE

A patient on a stretcher 'silent chest — impending failure'.

WHAT IT ENCODES

A 'silent chest' (absent breath sounds and absent wheeze) in an acutely dyspneic patient signals airflow so reduced that turbulent wheeze can no longer be generated — a sign of life-threatening exacerbation alongside cyanosis, exhaustion, altered consciousness, bradycardia, and a normalizing/rising PaCO₂.

BOARD TRAP

Loss of wheeze is OMINOUS, not reassuring — a quiet chest plus a 'normal' or rising pCO₂ in a tiring asthmatic means impending respiratory failure; prepare for ICU/intubation rather than de-escalating.

11. FEV down $\geq 10\%$ (variability)

WHAT YOU SEE

A water arch 'FEV down $\geq 10\%$ ' showing variability.

WHAT IT ENCODES

Documenting excessive variability in lung function over time — a fall in FEV₁ of $>10\%$ from baseline (e.g., after exercise challenge or between visits), or significant PEF swings — is an accepted objective criterion supporting the diagnosis when single-point reversibility is absent.

12. Mimics bench (HF / EGPA / VCD)

WHAT YOU SEE

A 'MIMICS BENCH' with heart failure, EGPA, and a runner with brown-plug/ABPA.

WHAT IT ENCODES

Asthma mimics that must be excluded: heart failure ('cardiac asthma' with orthopnea, edema, raised BNP), EGPA (asthma + sinusitis + marked eosinophilia + mononeuritis multiplex + ANCA), vocal-cord dysfunction (inspiratory stridor, flat inspiratory flow-volume loop), ABPA (brown mucus plugs, very high IgE, central bronchiectasis), COPD, and bronchiectasis.

BOARD TRAP

Asthma + chronic sinusitis + high eosinophils + peripheral neuropathy = EGPA (Churg-Strauss), not plain asthma — and a paradoxical worsening after starting/withdrawing leukotriene modifiers or steroid taper can unmask it.